

Lecture 01: Cloud Computing

Question Bank with Answers

Part 1: Multiple Choice (25)

1. What is the best description of cloud computing?

ما أفضل وصف للحوسبة السحابية؟

- A. Installing all software only on a local personal computer
- B. An on-demand delivery of IT capabilities and applications over the Internet
- C. Buying and owning all hardware inside the organization
- D. A network cable standard for connecting servers

✓ **Correct answer: B. An on-demand delivery of IT capabilities and applications over the Internet**

☞ له صح: الحوسبة السحابية هي تقديم قدرات تقنية المعلومات والتطبيقات عند الطلب عبر الإنترنت. الاختيارات الباقية تصف الحوسبة المحلية (On-Premises) وليس

☞ Ref: Lecture 01 — Slide 3

2. Which of the following is given as an example of a cloud solution?

أي مما يلي ذكر كمثال على حل سحابي؟

- A. A word processor installed offline on a laptop
- B. A local hard disk drive
- C. A USB flash memory
- D. Dropbox

✓ **Correct answer: D. Dropbox**

☞ له صح: المحاضرة ذكرت Gmail وFacebook وDropbox كأتملة سحابية. الباقي أجهزة أو برامج محلية لا تعتمد على الإنترنت.

☞ Ref: Lecture 01 — Slide 3

3. IT infrastructure and applications are provided to subscribers as _____ services over networks.

للتحتية والتطبيقات تُقدَّم للمشتركين كخدمات _____ عبر الشبكات.

- A. metered
- B. free-of-charge
- C. physical
- D. offline

✓ **Correct answer: A. metered**

☞ له صح: المحاضرة تنص على أنها خدمات مُقاسة (Metered)، أي يُحتسب استخدامها. ليست مجانية ولا فعلية ولا بدون اتصال.

☞ Ref: Lecture 01 — Slide 3

4. Which characteristic is also called scalability?

مع (scalability)؟

- A. Resource pooling
- B. Measured service
- C. Rapid elasticity
- D. Broad network access

✓ **Correct answer: C. Rapid elasticity**

☞ له صح: Rapid elasticity مكتوبة في المحاضرة بجانب Resource pooling، أي تجمع الموارد. Measured service قياس الاستخدام.

☞ Ref: Lecture 01 — Slide 4

5. Which characteristic means the provider combines its computing resources to serve multiple consumers?

أي خاصية تعني أن المزود يجمع موارده لخدمة عدة مستهلكين؟

- A. Resource pooling
- B. On-demand self-service
- C. Distributed storage
- D. Automated management

✓ **Correct answer: A. Resource pooling**

☞ له صح: Resource pooling هي تجميع الموارد لخدمة عدد من المستخدمين. Distributed storage تخص توزيع التخزين فقط.

☞ Ref: Lecture 01 — Slide 4

6. Which of the following is a challenge of cloud computing?

أي مما يلي يُعد من تحديات الحوسبة السحابية؟

- A. Unlimited control over the provider's hardware
- B. Difficulty in migrating from one service provider to another
- C. No dependence on network connections
- D. No security or privacy concerns

✓ **Correct answer: B. Difficulty in migrating from one service provider to another**

🔗 له صخ: صعوبة الانتقال بين مزودي الخدمة (Lock-in) من التحديات. المستخدم لديه تحكم محدود، ويعتمد على الشبكة، وتوجد مخاوف أمان وخصوصية.

📖 Ref: Lecture 01 — Slide 5

7. Why is a cloud potentially more vulnerable to attacks, according to the lecture?

لماذا تكون السحابة معرّضة للهجمات حسب المحاضرة؟

- A. Because it uses no virtualization
- B. Because data is never stored on servers
- C. Because every component is online
- D. Because it has no service providers

✓ **Correct answer: C. Because every component is online**

مكونات السحابة متصلة بالإنترنت فتزداد فرص الهجوم. باقي الاختيارات غير صحيحة ولا علاقة لها بالسبب.

📖 Ref: Lecture 01 — Slide 5

8. Which service enables subscribers to use on-demand computing power, virtualization, data storage, network and OS?

أي خدمة تتيح استخدام قدرة الحوسبة والافتراضية والتخزين والشبكة ونظام التشغيل عند الطلب؟

- A. Software-as-a-Service (SaaS)
- B. Identity-as-a-Service (IDaaS)
- C. Anything-as-a-Service (XaaS)
- D. Infrastructure-as-a-Service (IaaS)

✓ **Correct answer: D. Infrastructure-as-a-Service (IaaS)**

🔗 له صخ: IaaS تقدّم الموارد الأساسية لتقنية نظام التشغيل، التخزين، الشبكة، افتراضية، قدرة الحوسبة، استخدام عند الطلب.

📖 Ref: Lecture 01 — Slide 7

9. In PaaS, subscribers need NOT buy and manage the underlying software and infrastructure, but they have authority over:

في PaaS لا يحتاج المشترك لشراء وإدارة البرمجيات والبنية التحتية الأساسية، لكنه يتحكم في:

- A. The deployed applications
- B. The physical servers
- C. The virtualization layer
- D. The network cables

✓ **Correct answer: A. The deployed applications**

🔗 له صخ: PaaS يتيح تطوير التطبيقات مع بقاء السيطرة للمشارك على التطبيقات المنشورة، بينما المزود يدير الباقي.

📖 Ref: Lecture 01 — Slide 7

10. Which service offers application software to subscribers on-demand over the Internet?

أي خدمة تقدّم برامج تطبيقية للمشاركين عند الطلب عبر الإنترنت؟

- A. Platform-as-a-Service (PaaS)
- B. Infrastructure-as-a-Service (IaaS)
- C. Software-as-a-Service (SaaS)
- D. Function-as-a-Service (FaaS)

✓ **Correct answer: C. Software-as-a-Service (SaaS)**

🔗 له صخ: SaaS تقدّم البرنامج نفسه جاهزًا (مثل PaaS). (Google Docs و IaaS للبنية التحتية).

📖 Ref: Lecture 01 — Slide 7

11. Which service provides Single-Sign-On (SSO) and Multi-Factor Authentication (MFA)?

حدد (SSO) والتحقق متعدد العوامل (MFA)؟

- A. Container-as-a-Service (CaaS)
- B. Identity-as-a-Service (IDaaS)
- C. Function-as-a-Service (FaaS)
- D. Platform-as-a-Service (PaaS)

✓ Correct answer: B. Identity-as-a-Service (IDaaS)

🔗 له صح: IDaaS تدير الهوية والوصول عبر طرف ثالث وتشمل MFA وSSO. الخدمات الأخرى لا تركز على المصادقة.

📖 Ref: Lecture 01 — Slide 8

12. Security-as-a-Service (SECaaS) is developed based on which service type?

مبنى (SECaaS)؟

- A. IaaS
- B. FaaS
- C. CaaS
- D. SaaS

✓ Correct answer: D. SaaS

🔗 له صح: المحاضرة توضح أن SECaaS مبنية على SaaS، وتقلل التكلفة مقارنة ببناء قدرات أمنية خاصة بالمؤسسة.

📖 Ref: Lecture 01 — Slide 8

13. Which of the following services is provided by SECaaS?

بها (SECaaS)؟

- A. Penetration testing and intrusion detection
- B. Hosting food delivery services
- C. Running serverless functions only
- D. Providing virtual servers only

✓ Correct answer: A. Penetration testing and intrusion detection

🔗 له صح: SECaaS تقدم اختبار الاختراق والمصادقة وكشف التسلل ومكافحة البرمجيات الخبيثة. الباقي يخص IaaS وFaaS وXaaS.

📖 Ref: Lecture 01 — Slide 8

14. CaaS inherits features of which two service types?

تأخذ (CaaS)؟

- A. SaaS and IDaaS
- B. FaaS and XaaS
- C. IaaS and PaaS
- D. SECaaS and SaaS

✓ Correct answer: C. IaaS and PaaS

🔗 له صح: المحاضرة تذكر أن CaaS ترث خصائص IaaS وPaaS، وتستخدم لتطوير تطبيقات حاويات قابلة للتوسع.

📖 Ref: Lecture 01 — Slide 9

15. Which service is associated with serverless architecture?

م (serverless)؟

- A. Infrastructure-as-a-Service (IaaS)
- B. Function-as-a-Service (FaaS)
- C. Software-as-a-Service (SaaS)
- D. Identity-as-a-Service (IDaaS)

✓ Correct answer: B. Function-as-a-Service (FaaS)

🔗 له صح: FaaS تتيح تطوير وتشغيل وظائف التطبيق دون تعقيد بناء البنية التحتية وصيانتها.

📖 Ref: Lecture 01 — Slide 9

16. Which of the following is an example of XaaS offered over the Internet?

أي XaaS؟

- A. A local file server
- B. A physical router
- C. Medical consultations
- D. A desktop operating system license

✓ Correct answer: C. Medical consultations

☞ له صح: XaaS تقدم أي شيء كخدمة مثل الطعام والمواصلات والاستشارات الطبية.

📌 Ref: Lecture 01 — Slide 9

17. In the IaaS responsibility model, which layers does the subscriber manage?

المسؤولية؟

- A. Servers, storage and networking
- B. Virtualization and servers only
- C. Only networking
- D. Applications, data, runtime, middleware and OS

✓ Correct answer: D. Applications, data, runtime, middleware and OS

☞ له صح: في IaaS يدير المشترك من التطبيقات حتى نظام التشغيل، والمزود يدير الافتراضية والخوادم والتخزين والشبكات.

📌 Ref: Lecture 01 — Slide 10

18. In the PaaS responsibility model, the subscriber is responsible for:

لـ PaaS، المشترك مسؤول عن:

- A. Applications and data
- B. Runtime and middleware
- C. The operating system
- D. Servers and storage

✓ Correct answer: A. Applications and data

☞ له صح: في PaaS يتولى المزود كل شيء من Runtime إلى الشبكات، وتبقى مسؤولية المشترك على التطبيقات والبيانات فقط.

📌 Ref: Lecture 01 — Slide 10

19. Which cloud deployment model is for a single organization only?

أي نموذج نشر سحابي مخصص لمؤسسة واحدة فقط؟

- A. Public Cloud
- B. Community Cloud
- C. Private Cloud
- D. Hybrid Cloud

✓ Correct answer: C. Private Cloud

☞ له صح: Private Cloud لمؤسسة واحدة فقط، وCommunity Cloud لعدة مؤسسات من مجتمع محدد.

📌 Ref: Lecture 01 — Slide 11

20. A combination of two or more clouds (private, community, or public) is called:

يُتّين أو أكثر (خاصة أو مجتمعية أو عامة) يسمى:

- A. Public Cloud
- B. Hybrid Cloud
- C. Private Cloud
- D. Community Cloud

✓ Correct answer: B. Hybrid Cloud

☞ له صح: Hybrid Cloud هي مزيج من نموذجين أو أكثر، بينما الباقي نماذج منفردة.

📌 Ref: Lecture 01 — Slide 11

21. Which actor sets up service contracts with the Cloud Service Provider (CSP)?

بنة (CSP)؟

- A. Cloud Auditor
- B. Cloud Carrier
- C. Cloud Broker
- D. Cloud Consumer

✓ Correct answer: D. Cloud Consumer

☞ له صح: Cloud Consumer هو الشخص أو المؤسسة المستخدمة للخدمة، وهو من يبرم العقود مع CSP.

🔗 Ref: Lecture 01 — Slide 12

22. Which actor is an independent entity that evaluates and verifies cloud services, performance and security controls?

أي طرف مستقل يقيّم ويتحقق من الخدمات السحابية والأداء وضوابط الأمن؟

- A. Cloud Auditor
- B. Cloud Broker
- C. Cloud Provider
- D. Cloud Carrier

✓ Correct answer: A. Cloud Auditor

☞ له صح: Cloud Auditor يقيّم ويتحقق. Broker يدير العلاقات بين CSP والعميل. Carrier يوفر الاتصال والنقل.

🔗 Ref: Lecture 01 — Slide 12

23. Which actor provides connectivity and transport services between CSPs and cloud consumers via a network?

بين CSP؟

- A. Cloud Consumer
- B. Cloud Auditor
- C. Cloud Carrier
- D. Cloud Broker

✓ Correct answer: C. Cloud Carrier

☞ له صح: Cloud Carrier وسيط للاتصال والنقل. Broker يدير الاستخدام والأداء والتسليم والعلاقة.

🔗 Ref: Lecture 01 — Slide 12

24. Which situation is listed as a data breach/loss threat?

أي موقف مذكور كتهديد لاختراق أو فقدان البيانات؟

- A. Using a load balancer
- B. Encryption keys are lost
- C. Monitoring Quality of Service
- D. Implementing a disaster recovery plan

✓ Correct answer: B. Encryption keys are lost

دان مفاتيح التشفير أو حذف أو تعديل البيانات من أمثلة التهديد. الباقي إجراءات حماية.

🔗 Ref: Lecture 01 — Slide 14

25. According to the lecture, which role designs high-level system blueprints and selects frameworks and cloud services?

لمحاضرة، أي وظيفة تصمم المخططات عالية المستوى وتختار الأطر والخدمات السحابية؟

- A. Cloud Engineer
- B. Cloud Security Engineer
- C. DevOps Engineer
- D. Cloud Architect

✓ Correct answer: D. Cloud Architect

☞ له صح: Cloud Architect يصمم ويختار. DevOps Engineer و Security Engineer يضبط الوصول والتشفير.

🔗 Ref: Lecture 01 — Slide 17

Part 2: Essay (10)

1. Define cloud computing and give examples.

ف الحوسبة السحابية واذكر أمثلة.

Model answer:

- Cloud computing delivers various types of services and applications over the Internet.
- It is an on-demand delivery of IT capabilities.
- IT infrastructure and applications are provided to subscribers as metered services over networks.
- Examples: Gmail, Facebook, Dropbox.

➤ Ref: Lecture 01 — Slide 3

2. List and briefly explain at least five characteristics of cloud computing.

اشرح باختصار خمس خصائص على الأقل للحوسبة السحابية.

Model answer:

- On-demand self-service.
- Distributed storage.
- Rapid elasticity (scalability).
- Automated management.
- Broad network access.
- Resource pooling.
- Measured service.
- Virtualization technology.

➤ Ref: Lecture 01 — Slide 4

3. Discuss the challenges of cloud computing and the main cloud threats.

تحديات الحوسبة السحابية وأهم تهديداتها.

Model answer:

- Challenges: limited control, security and privacy, dependence on network connections, vulnerability to attacks as every component is online, and difficulty migrating between providers.
- Threats: data breach/loss (data erased or modified, encryption keys lost), unsynchronized system clocks, loss of operational and security logs.
- Illegal access due to weak authentication and authorization, natural disasters, hardware failure, and lock-in.

➤ Ref: Lecture 01 — Slides 5, 14

4. Compare IaaS, PaaS and SaaS.

بين IaaS وPaaS وSaaS.

Model answer:

- IaaS: on-demand fundamental IT resources such as computing power, virtualization, storage, network and OS.
- PaaS: allows development of applications and services; subscribers do not manage the underlying software and infrastructure but control deployed applications.
- SaaS: offers application software to subscribers on-demand over the Internet (e.g., Google Docs).

➤ Ref: Lecture 01 — Slide 7

5. Explain Identity-as-a-Service (IDaaS) and Security-as-a-Service (SECaaS).

ح IDaaS وSECaaS.

Model answer:

- IDaaS offers authentication services managed by a third-party vendor, providing identity and access management such as SSO and MFA.
- SECaaS is developed based on SaaS.
- SECaaS drastically reduces cost compared to building own security capabilities.
- It provides penetration testing, authentication, intrusion detection, anti-malware, etc.

➤ Ref: Lecture 01 — Slide 8

6. Explain CaaS, FaaS and XaaS.

ح CaaS وFaaS وXaaS.

Model answer:

- CaaS: lets subscribers develop rich scalable containerized applications; inherits features of both IaaS and PaaS.
- FaaS: a platform for developing, running and managing application functionalities without building and maintaining infrastructure (serverless); provides data processing services such as IoT.
- XaaS: other services such as food, transportation and medical consultations, and secure services such as customer relationship management.

🔗 Ref: Lecture 01 — Slide 9

7. Explain the division of responsibilities between subscriber and provider in On-Premises, IaaS, PaaS and SaaS.

ي On-Premises وIaaS وPaaS وSaaS.

Model answer:

- On-Premises: the resource owner manages all layers, from applications down to networking.
- IaaS: subscriber manages applications, data, runtime, middleware and OS; provider manages virtualization, servers, storage and networking.
- PaaS: subscriber manages applications and data; provider manages the rest.
- SaaS: the provider manages all layers.

🔗 Ref: Lecture 01 — Slide 10

8. Describe the four cloud deployment models.

نماذج النشر السحابية الأربعة.

Model answer:

- Selection is based on enterprise requirements.
- Public Cloud: open for public use.
- Private Cloud: for a single organization only.
- Community Cloud: several organizations from a specific community.
- Hybrid Cloud: combination of two or more clouds (private, community or public).

🔗 Ref: Lecture 01 — Slide 11

9. Name the five major actors in cloud computing and describe each.

الأطراف الخمسة الرئيسية في الحوسبة السحابية وشرح كل واحد.

Model answer:

- Cloud Consumer: uses cloud services and sets up contracts with the CSP.
- Cloud Provider: acquires and manages the computing infrastructure used to provide services.
- Cloud Carrier: intermediary providing connectivity and transport between CSPs and consumers.
- Cloud Auditor: independent entity evaluating and verifying services, performance and security controls.
- Cloud Broker: manages use, performance and delivery of services and the relationship between CSPs and consumers.

🔗 Ref: Lecture 01 — Slide 12

10. Explain cloud security as a shared responsibility and list the main security considerations.

مسئولية مشتركة، واذكر أهم اعتبارات الأمن.

Model answer:

- Cloud security is shared between the provider (security OF the cloud) and the customer (security IN the cloud).
- Implement a disaster recovery plan.
- Continuously monitor QoS to maintain service level agreements.
- Ensure data integrity for stored data.
- Services should be fast, reliable, with quick response times.
- Use cryptographic algorithms and incorporate load balancing.

🔗 Ref: Lecture 01 — Slides 15, 16